



User: Jakub Murcek
Project: Master Thesis

```
1 .
2 . * Merge core_ai into applied_ai (only 6 obs - below 50-obs threshold)
3 . replace desc_tier_llm = "applied_ai" if desc_tier_llm == "core_ai"
  (6 real changes made)

4 .
5 . encode desc_tier_llm, generate(ai_tier_num)

6 .
7 . * --- 3.2 AI Flag (Strict Intersection + Buzzword Filter) ---
8 . * 1. Sloučime zdroje skills do jednoho retezce pro kontrolu
9 . gen skills_combined = lower(skills_ai_det + " " + desc_ai_llm)

10 .
11 . * 2. Odstranime obecne buzzwords (AI, ML, Artificial Intelligence atd.)
12 . * Pouzivame regex s word boundaries \b aby se smazalo jen "AI" a ne "OpenAI"
13 . gen skills_no_buzz = ustrregexra(skills_combined, "(?i)\b(ai|ml|artificial intelligence|machine learning|genai)\b", "")

14 .
15 . * 3. Odstranime interpunkci (carky, mezery), abychom zarucili ze zbyva neco realneho
16 . gen skills_cleaned = subinstr(skills_no_buzz, ",", "", .)

17 . replace skills_cleaned = subinstr(skills_cleaned, " ", "", .)
  (17,848 real changes made)

18 . replace skills_cleaned = subinstr(skills_cleaned, ";", "", .)
  (0 real changes made)

19 .
20 . * 4. Definice has_ai_flag
21 . * Tier != None A ZAROVEN zbyly nejake specificke skills (delka > 1 znaku)
22 . gen has_ai_flag = ((desc_tier_llm != "none" & desc_tier_llm != "missing") & length(skills_cleaned) > 1)

23 .
```

```
24 . label variable has_ai_flag "AI Job (Tier + Valid Skills, no buzzwords)"

25 .
26 . * Kompatibilita pro starší skripty (volitelné)
27 . gen has_ai = has_ai_flag

28 .
29 . * --- 3.3 Vzdělání (Hybridní) ---
30 . * Vytvoříme 'education_hybrid' z edulevel_llm (primární) a edu_level_det (fallback)
31 . * Výsledné hodnoty: "missing", "highschool", "associate", "bachelor", "master", "phd"
32 . * POZOR: encode řadí abecedně, proto kódujeme manuálně (ordinální pořadí)
33 .
34 . * Krok 1: Normalizujeme edulevel_llm na lowercase a sjednotíme hodnoty
35 . gen education_hybrid = lower(edulevel_llm)

36 . replace education_hybrid = subinstr(education_hybrid, "'s", "", .)
    (9,186 real changes made)

37 . replace education_hybrid = "highschool" if education_hybrid == "high school"
    (312 real changes made)

38 . replace education_hybrid = "missing" if education_hybrid == "-" | education_hybrid == ""
    (8,071 real changes made)

39 .
40 . * Krok 2: Fallback na edu_level_det pokud LLM chybí
41 . replace education_hybrid = edu_level_det if education_hybrid == "missing" & edu_level_det != ""
    (1,707 real changes made)

42 .
43 . * Krok 3: Finální missing pro prázdné
44 . replace education_hybrid = "missing" if education_hybrid == ""
    (0 real changes made)

45 .
```

```
46 . gen edu_cat = .  
    (17,848 missing values generated)  
  
47 . replace edu_cat = 0 if inlist(education_hybrid, "missing", "", "highschool", "associate")  
    (7,077 real changes made)  
  
48 . replace edu_cat = 1 if inlist(education_hybrid, "bachelor", "master", "phd")  
    (10,771 real changes made)  
  
49 .  
50 . label define edu_lbl 0 "No Degree / Missing" 1 "Bachelor or Higher"  
  
51 . label values edu_cat edu_lbl  
  
52 . label variable edu_cat "Pozadovane vzdelani (Binary)"  
  
53 .  
54 . * --- 3.4 Zkušenosti ---  
55 . * experience_min_llm by měl být již numerický, ale pro jistotu  
56 . destring experience_min_llm, replace force  
    experience_min_llm: contains nonnumeric characters; replaced as double  
    (2591 missing values generated)  
  
57 . label variable experience_min_llm "Min. pozadovane roky zkusenosti"  
  
58 .  
59 . * Kategorie zkušeností  
60 . * Missing = neuvedeno v inzerátu (samostatná kategorie, jako u vzdělání)  
61 . gen exp_category = .  
    (17,848 missing values generated)  
  
62 . replace exp_category = 0 if experience_min_llm == .  
    (2,591 real changes made)  
  
63 . * Sloučení Entry s Junior a Expert se Senior
```

```
64 . replace exp_category = 2 if experience_min_llm >= 0 & experience_min_llm <= 2
    (3,213 real changes made)

65 . replace exp_category = 3 if experience_min_llm > 2 & experience_min_llm <= 5
    (8,406 real changes made)

66 . replace exp_category = 4 if experience_min_llm > 5 & experience_min_llm < .
    (3,638 real changes made)

67 .
68 . label define exp_lbl 0 "Missing" 2 "Junior (0-2)" 3 "Mid (3-5)" 4 "Senior+ (6+)"

69 . label values exp_category exp_lbl

70 . label variable exp_category "Kategorie seniority"

71 .
72 . * --- 3.5 Plat ---
73 . * Převod na roční plat podle pay_period
74 . * US standard: 2080 hodin/rok (40h/týden × 52 týdnů), 12 měsíců/rok
75 . destring salary_min salary_mid salary_max, replace force
    salary_min already numeric; no replace
    salary_mid already numeric; no replace
    salary_max already numeric; no replace

76 .
77 . * Přepočet hodinových mezd na roční (HOURLY × 2080)
78 . replace salary_min = salary_min * 2080 if pay_period == "HOURLY"
    (840 real changes made)

79 . replace salary_mid = salary_mid * 2080 if pay_period == "HOURLY"
    (840 real changes made)

80 . replace salary_max = salary_max * 2080 if pay_period == "HOURLY"
    (840 real changes made)
```

```
81 .
82 . * Přepočet měsíčních mezd na roční (MONTHLY × 12)
83 . replace salary_min = salary_min * 12 if pay_period == "MONTHLY"
    (67 real changes made)

84 . replace salary_mid = salary_mid * 12 if pay_period == "MONTHLY"
    (67 real changes made)

85 . replace salary_max = salary_max * 12 if pay_period == "MONTHLY"
    (67 real changes made)

86 .
87 . * Filtr outlierů (po přepočtu na roční bázi)
88 . replace salary_mid = . if salary_mid < 20000 | salary_mid > 500000
    (5 real changes made, 5 to missing)

89 . label variable salary_mid "Rocni plat - stredni hodnota (USD)"

90 .
91 . * --- 3.6 Sektor a Lokace ---
92 . replace sector = "Unknown" if sector == ""
    (3,500 real changes made)

93 . replace industry = "Unknown" if industry == ""
    (3,501 real changes made)

94 . encode sector, generate(sector_num)

95 .
96 . replace state = "Unknown" if state == ""
    (4,547 real changes made)

97 . encode state, generate(state_num)

98 .
```

```
99 . * --- 3.8 Velikost firmy (ordinální) ---
100 . replace size = "Unknown" if size == ""
    (860 real changes made)

101 . gen size_cat = .
    (17,848 missing values generated)

102 . replace size_cat = 0 if size == "Unknown"
    (2,703 real changes made)

103 . replace size_cat = 1 if size == "1 to 50 Employees"
    (3,152 real changes made)

104 . replace size_cat = 2 if size == "51 to 200 Employees"
    (2,324 real changes made)

105 . replace size_cat = 3 if size == "201 to 500 Employees"
    (1,580 real changes made)

106 . replace size_cat = 4 if size == "501 to 1000 Employees"
    (1,051 real changes made)

107 . replace size_cat = 5 if size == "1001 to 5000 Employees"
    (2,237 real changes made)

108 . replace size_cat = 6 if size == "5001 to 10000 Employees"
    (1,074 real changes made)

109 . replace size_cat = 7 if size == "10000+ Employees"
    (3,727 real changes made)

110 .
111 . label define size_lbl 0 "Unknown" 1 "1-50" 2 "51-200" 3 "201-500" ///
    >      4 "501-1000" 5 "1001-5000" 6 "5001-10000" 7 "10000+"

```

```
112 . label values size_cat size_lbl
113 . label variable size_cat "Velikost firmy (ordinalni)"
114 .
115 . * --- 3.9 Typ firmy (nominální, sloučený) ---
116 . gen type_cat = .
    (17,848 missing values generated)
117 . replace type_cat = 0 if inlist(type, "", "Unknown", "Contract", "Self-employed", "Private Practice / Firm", "Franchise") // Merge Other to
    (1,504 real changes made)
118 . replace type_cat = 1 if inlist(type, "Company - Private", "Subsidiary or Business Segment") // Merge Subsidiary to Private
    (9,733 real changes made)
119 . replace type_cat = 2 if type == "Company - Public"
    (5,707 real changes made)
120 . replace type_cat = 4 if inlist(type, "Nonprofit Organization", "Government", ///
    > "College / University", "School / School District", "Hospital")
    (904 real changes made)
121 .
122 . label define type_lbl 0 "Unknown/Other" 1 "Private/Subsidiary" 2 "Public" ///
    > 4 "Nonprofit/Gov/Edu"
123 . label values type_cat type_lbl
124 . label variable type_cat "Typ firmy"
125 .
126 . * --- 3.10 NACE sektor (přichází z clean-stata CLI, jen encode) ---
127 . replace sector_nace = "Unknown" if sector_nace == ""
    (0 real changes made)
```

```
128 .
129 . * Zachovat top 5 sektorů + Unknown, vše ostatní sloučit do "Other" (kvůli < 50 obs v Applied AI)
130 . replace sector_nace = "Other" if !inlist(sector_nace, "J", "C", "K", "M", "Q", "Unknown")
    (2,281 real changes made)

131 .
132 . encode sector_nace, generate(sector_nace_num)

133 . label variable sector_nace_num "NACE sektor"

134 .
135 . * --- 3.11 Rok založení firmy ---
136 . destring year_founded, replace force
    year_founded already numeric; no replace

137 . label variable year_founded "Rok zalozeni firmy"

138 .
139 . * --- 3.7 Remote práce ---
140 . gen is_remote = 0

141 . replace is_remote = 1 if strpos(lower(remote_work_types), "home") > 0
    (5,067 real changes made)

142 . replace is_remote = 1 if strpos(lower(remote_work_types), "remote") > 0
    (0 real changes made)

143 .
144 . label variable is_remote "Moznost remote prace (1=ano)"

145 .
146 . * --- 3.12 AI Level (multinomiální závislá proměnná) ---
147 . * Tříkategorická proměnná pro multinomiální logit:
148 . *   0 = žádné AI požadavky (none)
149 . *   1 = používá AI nástroje (ai_integration)
```

```
150 . * 2 = vyvíjí/buduje AI (applied_ai - core_ai already merged above)
151 . gen ai_level = 0

152 . replace ai_level = 1 if desc_tier_llm == "ai_integration"
    (2,354 real changes made)

153 . replace ai_level = 2 if desc_tier_llm == "applied_ai"
    (1,313 real changes made)

154 .
155 . label define ailevel_lbl 0 "None" 1 "AI Integration" 2 "Applied/Core AI"

156 . label values ai_level ailevel_lbl

157 . label variable ai_level "Uroveň AI požadavku (0/1/2)"

158 .
159 . * --- 3.13 Census Region (přichází z clean-stata CLI, jen encode) ---
160 . replace region = "Unknown" if region == ""
    (0 real changes made)

161 . encode region, generate(region_num)

162 . label variable region_num "Census region"

163 .
164 . * --- 3.14 Job Family (přichází z clean-stata CLI, jen encode) ---
165 . replace job_family = "Unknown" if job_family == ""
    (0 real changes made)

166 .
167 . * Sloučení řídkých technických rolí do "Other" (kvůli < 50 obs v Applied AI)
168 . replace job_family = "Other" if inlist(job_family, "Frontend & Design", "QA & Testing", "Security", "Systems & Embedded")
    (5,005 real changes made)
```

```

169 .
170 . encode job_family, generate(job_family_num)

171 . label variable job_family_num "Rodina pozice"

172 .
173 . * --- 3.15 Vyřazení neplatných skill clusterů ---
174 . * Tyto clustery mají v kategorii Applied/Core AI méně než 50 pozorování, což by narušilo model.
175 . * Proto je mažeme, aby je nezahrnoval wildcard cluster_* v regresních modelech.
176 . drop cluster_legacy__mainframe

177 . drop cluster_data_analysis__stats

178 . drop cluster_tools__editors

179 .
180 . * =====
181 . * 4. DESKRIPTIVNÍ STATISTIKA
182 . * =====
183 . * Základní přehled datasetu - toto jde typicky do první tabulky v diplomce
184 .
185 . display _n "=====
=====

186 . display "4. DESKRIPTIVNI STATISTIKA"
4. DESKRIPTIVNI STATISTIKA

187 . display "=====
=====

188 .
189 . * --- 4.1 Frekvence AI tier klasifikace ---
190 . * KLÍČOVÁ TABULKA: Kolik % pozic vyžaduje AI dovednosti?
191 . display _n "--- 4.1 Distribuce AI pozadavku v IT pozicich ---"

--- 4.1 Distribuce AI pozadavku v IT pozicich ---

```

```
192 . tab desc_tier_llm, missing
```

desc_tier_llm	Freq.	Percent	Cum.
ai_integration	2,354	13.19	13.19
applied_ai	1,313	7.36	20.55
none	14,181	79.45	100.00
Total	17,848	100.00	

```
193 . tab desc_tier_llm if desc_tier_llm != "missing", sort
```

desc_tier_llm	Freq.	Percent	Cum.
none	14,181	79.45	79.45
ai_integration	2,354	13.19	92.64
applied_ai	1,313	7.36	100.00
Total	17,848	100.00	

```
194 .
```

```
195 . * Procentuální rozdělení (pro text diplomky)
```

```
196 . count if has_ai == 1
```

```
3,477
```

```
197 . local n_ai = r(N)
```

```
198 . count
```

```
17,848
```

```
199 . local n_total = r(N)
```

```
200 . display _n "Podíl pozic s AI požadavky: " %5.2f (`n_ai'/'n_total')*100 "%"
```

```
Podíl pozic s AI požadavky: 19.48%
```

```

201 .
202 . * --- 4.2 Požadované vzdělání ---
203 . display _n "--- 4.2 Distribuce vzdelavacich pozadavku ---"

```

--- 4.2 Distribuce vzdelavacich pozadavku ---

```
204 . tab edu_cat, missing
```

Pozadovane vzdelani (Binary)	Freq.	Percent	Cum.
No Degree / Missing	7,077	39.65	39.65
Bachelor or Higher	10,771	60.35	100.00
Total	17,848	100.00	

```
205 . tab edu_cat if edu_cat > 0
```

Pozadovane vzdelani (Binary)	Freq.	Percent	Cum.
Bachelor or Higher	10,771	100.00	100.00
Total	10,771	100.00	

```

206 .
207 . * Porovnání vzdělání podle AI tier
208 . display _n "Vzdelani x AI tier (kontingencni tabulka):"

```

Vzdelani x AI tier (kontingencni tabulka):

```
209 . tab edu_cat ai_tier_num if edu_cat > 0, chi2 column
```

Key
<i>frequency</i> <i>column percentage</i>

Pozadovane vzdelani (Binary)	ai_tier_num			Total
	ai_integr	applied_a	none	
Bachelor or Higher	1,243 100.00	756 100.00	8,772 100.00	10,771 100.00
Total	1,243 100.00	756 100.00	8,772 100.00	10,771 100.00

```
210 .
211 . * --- 4.3 Požadované zkušenosti ---
212 . display _n "--- 4.3 Distribuce pozadavku na zkusenosti ---"
```

--- 4.3 Distribuce pozadavku na zkusenosti ---

```
213 . summarize experience_min_llm, detail
```

Min. pozadovane roky zkusenosti				
Percentiles		Smallest		
1%	0	0		
5%	.5	0		
10%	1	0	Obs	15,257
25%	3	0	Sum of Wgt.	15,257
50%	5	Largest	Mean	4.520433
75%	5		Std. Dev.	2.625669
90%	8	25	Variance	6.894137
95%	10	25	Skewness	.9726134
99%	12	35	Kurtosis	6.37971

```
214 .
215 . tab exp_category, missing
```

Kategorie seniority	Freq.	Percent	Cum.
Missing	2,591	14.52	14.52
Junior (0-2)	3,213	18.00	32.52
Mid (3-5)	8,406	47.10	79.62
Senior+ (6+)	3,638	20.38	100.00
Total	17,848	100.00	

```
216 . tab exp_category ai_tier_num, chi2 column
```

Key
<i>frequency</i>
<i>column percentage</i>

Kategorie seniority	ai_integr	ai_tier_num applied_a	none	Total
Missing	265 11.26	177 13.48	2,149 15.15	2,591 14.52
Junior (0-2)	352 14.95	200 15.23	2,661 18.76	3,213 18.00
Mid (3-5)	1,238 52.59	620 47.22	6,548 46.17	8,406 47.10
Senior+ (6+)	499 21.20	316 24.07	2,823 19.91	3,638 20.38
Total	2,354 100.00	1,313 100.00	14,181 100.00	17,848 100.00

Pearson chi2(6) = **73.2360** Pr = **0.000**

```
217 .
218 . * --- 4.4 Platy ---
219 . display_n "---- 4.4 Distribuce platu ----"
```

--- 4.4 Distribuce platu ---

```
220 . summarize salary_mid, detail
```

Rocni plat - stredni hodnota (USD)

	Percentiles	Smallest		
1%	55000	20000		
5%	70000	29120		
10%	78944	30000	Obs	14,640
25%	95000	30000	Sum of Wgt.	14,640
50%	117884.5		Mean	124631.6
		Largest	Std. Dev.	42133.04
75%	147736.5	445000		
90%	180000	445000	Variance	1.78e+09
95%	200000	445000	Skewness	1.338601
99%	244250	450000	Kurtosis	7.674263

```
221 .
222 . * Platy podle AI tier - DŮLEŽITĚ pro argument o "AI premium"
223 . display _n "Plat podle AI tier:"
```

Plat podle AI tier:

```
224 . tabstat salary_mid, by(desc_tier_llm) statistics(count mean sd min p25 p50 p75 max)
```

Summary for variables: salary_mid
by categories of: desc_tier_llm

desc_tier_llm	N	mean	sd	min	p25	p50	p75	max
ai_integration	1946	140487.2	46483.01	31200	106500	135000	168000	410000
applied_ai	1094	150497.5	48356.8	46532	115403	148495	179000	445000
none	11600	119532.3	39070.47	20000	92406	114000	140000	450000
Total	14640	124631.6	42133.04	20000	95000	117884.5	147736.5	450000

```

225 .
226 . * --- 4.5 Sektory a industrie ---
227 . display _n "---- 4.5 Top 10 sektoru ----"

```

--- 4.5 Top 10 sektoru ---

```

228 . tab sector if sector != "Unknown", sort

```

sector	Freq.	Percent	Cum.
Information Technology	6,071	42.31	42.31
Manufacturing	1,219	8.50	50.81
Financial Services	1,081	7.53	58.34
Aerospace & Defense	981	6.84	65.18
Management & Consulting	707	4.93	70.11
Healthcare	692	4.82	74.93
Media & Communication	546	3.81	78.74
Retail & Wholesale	405	2.82	81.56
Education	375	2.61	84.17
Insurance	321	2.24	86.41
Human Resources & Staffing	294	2.05	88.46
Government & Public Administration	271	1.89	90.35
Pharmaceutical & Biotechnology	229	1.60	91.94
Energy, Mining & Utilities	184	1.28	93.23
Telecommunications	175	1.22	94.45
Construction, Repair & Maintenance Se..	170	1.18	95.63
Transportation & Logistics	143	1.00	96.63
Arts, Entertainment & Recreation	102	0.71	97.34
Real Estate	94	0.66	97.99
Nonprofit & NGO	76	0.53	98.52
Personal Consumer Services	56	0.39	98.91
Restaurants & Food Service	48	0.33	99.25
Legal	44	0.31	99.55
Hotels & Travel Accommodation	33	0.23	99.78
Agriculture	30	0.21	99.99
Media	1	0.01	100.00
Total	14,348	100.00	

```

229 .
230 . display _n "---- 4.6 Remote prace ---"

```

```

--- 4.6 Remote prace ---

```

```

231 . tab is_remote

```

Moznost remote prace (1=ano)	Freq.	Percent	Cum.
0	12,781	71.61	71.61
1	5,067	28.39	100.00
Total	17,848	100.00	

```

232 . tab is_remote has_ai, chi2 row

```

Key
<i>frequency</i> <i>row percentage</i>

Moznost remote prace (1=ano)	has_ai		Total
	0	1	
0	10,612 83.03	2,169 16.97	12,781 100.00
1	3,759 74.19	1,308 25.81	5,067 100.00
Total	14,371 80.52	3,477 19.48	17,848 100.00

Pearson chi2(1) = **180.9134** Pr = **0.000**

```

233 .
234 . * --- 4.7 Velikost firmy ---
235 . display _n "--- 4.7 Velikost firmy ---"

```

--- 4.7 Velikost firmy ---

```
236 . tab size_cat, missing
```

Velikost firmy (ordinalni)	Freq.	Percent	Cum.
Unknown	2,703	15.14	15.14
1-50	3,152	17.66	32.80
51-200	2,324	13.02	45.83
201-500	1,580	8.85	54.68
501-1000	1,051	5.89	60.57
1001-5000	2,237	12.53	73.10
5001-10000	1,074	6.02	79.12
10000+	3,727	20.88	100.00
Total	17,848	100.00	

```

237 .
238 . * --- 4.8 Typ firmy ---
239 . display _n "--- 4.8 Typ firmy ---"

```

--- 4.8 Typ firmy ---

```
240 . tab type_cat, missing
```

Typ firmy	Freq.	Percent	Cum.
Unknown/Other	1,504	8.43	8.43
Private/Subsidiary	9,733	54.53	62.96
Public	5,707	31.98	94.94
Nonprofit/Gov/Edu	904	5.06	100.00
Total	17,848	100.00	

```

241 .
242 . * --- 4.9 NACE sektor ---
243 . display _n "---- 4.9 NACE sektor ----"

```

--- 4.9 NACE sektor ---

```

244 . tab sector_nace if sector_nace != "Unknown", sort

```

sector_nace	Freq.	Percent	Cum.
J	6,793	47.34	47.34
C	2,429	16.93	64.27
Other	2,281	15.90	80.17
K	1,402	9.77	89.94
M	751	5.23	95.18
Q	692	4.82	100.00
Total	14,348	100.00	

```

245 .
246 . * --- 4.10 Rok zalozeni ---
247 . display _n "---- 4.10 Rok zalozeni firmy ----"

```

--- 4.10 Rok zalozeni firmy ---

```

248 . summarize year_founded, detail

```

Rok zalozeni firmy				
Percentiles		Smallest		
1%	1812	1625		
5%	1885	1630		
10%	1918	1636	Obs	11,719
25%	1972	1636	Sum of Wgt.	11,719
50%	1998	Largest	Mean	1981.815
			Std. Dev.	44.89752
75%	2010	2025		
90%	2016	2025	Variance	2015.787
95%	2019	2025	Skewness	-2.326764
99%	2022	2025	Kurtosis	10.05919

```

249 .
250 . * --- 4.11 Census Region ---
251 . display _n "---- 4.11 Census Region ----"

```

--- 4.11 Census Region ---

```
252 . tab region, missing
```

region	Freq.	Percent	Cum.
Midwest	1,932	10.82	10.82
Northeast	2,116	11.86	22.68
South	5,145	28.83	51.51
Unknown	4,601	25.78	77.29
West	4,054	22.71	100.00
Total	17,848	100.00	

```
253 . tab region has_ai, chi2 column
```

Key
<i>frequency</i>
<i>column percentage</i>

region	has_ai		Total
	0	1	
Midwest	1,659	273	1,932
	11.54	7.85	10.82
Northeast	1,735	381	2,116
	12.07	10.96	11.86
South	4,454	691	5,145
	30.99	19.87	28.83
Unknown	3,459	1,142	4,601
	24.07	32.84	25.78
West	3,064	990	4,054
	21.32	28.47	22.71
Total	14,371	3,477	17,848
	100.00	100.00	100.00

Pearson chi2(4) = 304.9573 Pr = 0.000

```
254 .
255 . * --- 4.12 Job Family ---
256 . display _n "--- 4.12 Job Family ---"
```

--- 4.12 Job Family ---

```
257 . tab job_family, missing
```

job_family	Freq.	Percent	Cum.
Data & AI	944	5.29	5.29
DevOps & Cloud	1,497	8.39	13.68
Management	786	4.40	18.08
Other	8,078	45.26	63.34
Software Developer	3,284	18.40	81.74
Software Engineer	2,039	11.42	93.16
Sr+ Software Engineer	1,220	6.84	100.00
Total	17,848	100.00	

```
258 . tab job_family has_ai, chi2 column
```

Key
<i>frequency</i>
<i>column percentage</i>

job_family	has_ai		Total
	0	1	
Data & AI	430 2.99	514 14.78	944 5.29
DevOps & Cloud	1,265 8.80	232 6.67	1,497 8.39
Management	543 3.78	243 6.99	786 4.40
Other	6,998 48.70	1,080 31.06	8,078 45.26

Software Developer	2,790 19.41	494 14.21	3,284 18.40
Software Engineer	1,514 10.54	525 15.10	2,039 11.42
Sr+ Software Engineer	831 5.78	389 11.19	1,220 6.84
Total	14,371 100.00	3,477 100.00	17,848 100.00

Pearson chi2(6) = **1.2e+03** Pr = **0.000**

```
259 .
260 . * --- 4.13 Skill Cluster frequencies ---
261 . display _n "--- 4.13 Pocet pozic s danym skill clusterem ---"
```

--- 4.13 Pocet pozic s danym skill clusterem ---

```
262 . foreach var of varlist cluster_* {
      2.   quietly count if `var' == 1
      3.   display "`var': " r(N) " jobs (" %4.1f r(N)/_N*100 "%) "
      4. }
cluster_architecture__methods: 14528 jobs (81.4%)
cluster_bi__analytics: 6131 jobs (34.4%)
cluster_backend_development: 9770 jobs (54.7%)
cluster_certifications: 1034 jobs ( 5.8%)
cluster_cloud_computing: 7714 jobs (43.2%)
cluster_data_engineering: 5543 jobs (31.1%)
cluster_data_science__ml: 2153 jobs (12.1%)
cluster_databases__storage: 8025 jobs (45.0%)
cluster_devops__containers: 8534 jobs (47.8%)
cluster_dynamic__web: 9942 jobs (55.7%)
cluster_enterprise__managed: 6339 jobs (35.5%)
cluster_enterprise_platforms: 830 jobs ( 4.7%)
cluster_frontend_development: 10759 jobs (60.3%)
cluster_generative_ai: 1224 jobs ( 6.9%)
cluster_mobile__desktop: 2962 jobs (16.6%)
cluster_networking: 4951 jobs (27.7%)
cluster_os__embedded: 3769 jobs (21.1%)
cluster_scripting__shell: 1983 jobs (11.1%)
cluster_security__identity: 5853 jobs (32.8%)
cluster_systems_programming: 2822 jobs (15.8%)
cluster_testing_qa__debugging: 9837 jobs (55.1%)
```

```

263 .
264 . * --- 4.14 Cross-tabs of model IVs with ai_level (check for empty cells) ---
265 . display _n "--- 4.14 Krizove tabulky pro kontrolu prazdnych bunek ---"

```

--- 4.14 Krizove tabulky pro kontrolu prazdnych bunek ---

```

266 . display _n "NACE sektor x AI level:"

```

NACE sektor x AI level:

```

267 . tab sector_nace ai_level, column

```

Key
<i>frequency</i>
<i>column percentage</i>

sector_nace	Uroven AI pozadavku (0/1/2)			Total
	None	AI Integr	Applied/C	
C	2,054	224	151	2,429
	14.48	9.52	11.50	13.61
J	5,258	971	564	6,793
	37.08	41.25	42.96	38.06
K	1,118	190	94	1,402
	7.88	8.07	7.16	7.86
M	586	107	58	751
	4.13	4.55	4.42	4.21
Other	1,894	261	126	2,281
	13.36	11.09	9.60	12.78
Q	528	113	51	692
	3.72	4.80	3.88	3.88
Unknown	2,743	488	269	3,500
	19.34	20.73	20.49	19.61
Total	14,181	2,354	1,313	17,848
	100.00	100.00	100.00	100.00

```
268 . display _n "Typ firmy x AI level:"
```

Typ firmy x AI level:

```
269 . tab type_cat ai_level, column
```

Key
<i>frequency</i>
<i>column percentage</i>

Typ firmy	Uroven AI pozadavku (0/1/2)			Total
	None	AI Integr	Applied/C	
Unknown/Other	1,184 8.35	211 8.96	109 8.30	1,504 8.43
Private/Subsidiary	7,721 54.45	1,291 54.84	721 54.91	9,733 54.53
Public	4,534 31.97	743 31.56	430 32.75	5,707 31.98
Nonprofit/Gov/Edu	742 5.23	109 4.63	53 4.04	904 5.06
Total	14,181 100.00	2,354 100.00	1,313 100.00	17,848 100.00

```
270 . display _n "Velikost firmy x AI level:"
```

Velikost firmy x AI level:

```
271 . tab size_cat ai_level, column
```

Key
<i>frequency</i>
<i>column percentage</i>

Velikost firmy (ordinalni)	Uroven AI pozadavku (0/1/2)			Total
	None	AI Integr	Applied/C	
Unknown	2,180 15.37	345 14.66	178 13.56	2,703 15.14
1-50	2,590 18.26	353 15.00	209 15.92	3,152 17.66
51-200	1,807 12.74	331 14.06	186 14.17	2,324 13.02
201-500	1,234 8.70	237 10.07	109 8.30	1,580 8.85
501-1000	828 5.84	155 6.58	68 5.18	1,051 5.89
1001-5000	1,728 12.19	340 14.44	169 12.87	2,237 12.53
5001-10000	883 6.23	122 5.18	69 5.26	1,074 6.02
10000+	2,931 20.67	471 20.01	325 24.75	3,727 20.88
Total	14,181 100.00	2,354 100.00	1,313 100.00	17,848 100.00

272 . display _n "Region x AI level:"

Region x AI level:

273 . tab region ai_level, column

Key
<i>frequency</i>
<i>column percentage</i>

region	Uroven AI pozadavku (0/1/2)			Total
	None	AI Integr	Applied/C	
Midwest	1,648 11.62	196 8.33	88 6.70	1,932 10.82
Northeast	1,712 12.07	260 11.05	144 10.97	2,116 11.86
South	4,411 31.10	481 20.43	253 19.27	5,145 28.83
Unknown	3,412 24.06	757 32.16	432 32.90	4,601 25.78
West	2,998 21.14	660 28.04	396 30.16	4,054 22.71
Total	14,181 100.00	2,354 100.00	1,313 100.00	17,848 100.00

274 . display _n "Job Family x AI level:"

Job Family x AI level:

275 . tab job_family ai_level, column

Key
<i>frequency</i>
<i>column percentage</i>

job_family	Uroven AI pozadavku (0/1/2)			Total
	None	AI Integr	Applied/C	
Data & AI	418 2.95	185 7.86	341 25.97	944 5.29
DevOps & Cloud	1,230 8.67	183 7.77	84 6.40	1,497 8.39
Management	536 3.78	136 5.78	114 8.68	786 4.40
Other	6,927 48.85	901 38.28	250 19.04	8,078 45.26
Software Developer	2,763 19.48	373 15.85	148 11.27	3,284 18.40
Software Engineer	1,493 10.53	327 13.89	219 16.68	2,039 11.42
Sr+ Software Engineer	814 5.74	249 10.58	157 11.96	1,220 6.84
Total	14,181 100.00	2,354 100.00	1,313 100.00	17,848 100.00

```

276 .
277 .
278 . * =====
279 . * 5. ANALYTICKÉ TESTY - HYPOTÉZY
280 . * =====
281 . * Statistické testy pro ověření hypotéz diplomové práce
282 .
283 . display _n "=====

```

```
=====
```

```

284 . display "5. STATISTICKE TESTY"
5. STATISTICKE TESTY

```

```

285 . display "=====
=====

```

```

286 .
287 . * --- 5.1 T-test: Liší se platy AI vs non-AI pozic? ---
288 . * H0: Průměrný plat AI pozic = průměrný plat non-AI pozic
289 . * H1: Průměrné platy se liší
290 .
291 . display _n "--- 5.1 T-test: Plat AI vs non-AI pozic ---"

```

--- 5.1 T-test: Plat AI vs non-AI pozic ---

```
292 . ttest salary_mid, by(has_ai)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
0	11,765	119901.6	363.3237	39408.48	119189.5	120613.8
1	2,875	143987.6	878.5001	47104.32	142265	145710.1
combined	14,640	124631.6	348.2188	42133.04	123949.1	125314.2
diff		-24085.93	853.6782		-25759.24	-22412.61

```

diff = mean(0) - mean(1)                                t = -28.2143
Ho: diff = 0                                             degrees of freedom = 14638

```

```

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000         Pr(|T| > |t|) = 0.0000         Pr(T > t) = 1.0000

```

```

293 .
294 . * Efekt size (Cohenovo d) - důležité pro interpretaci praktické významnosti
295 . quietly summarize salary_mid if has_ai == 0

296 . local mean_no_ai = r(mean)

297 . local sd_no_ai = r(sd)

298 . local n_no_ai = r(N)

299 .
300 . quietly summarize salary_mid if has_ai == 1

301 . local mean_ai = r(mean)

302 . local sd_ai = r(sd)

303 . local n_ai = r(N)

304 .
305 . * Pooled standard deviation
306 . local sd_pooled = sqrt(((`n_no_ai'-1)*`sd_no_ai'^2 + (`n_ai'-1)*`sd_ai'^2) / (`n_no_ai'+`n_ai'-2))

307 . local cohens_d = (`mean_ai' - `mean_no_ai') / `sd_pooled'

308 .
309 . display _n "Cohenovo d (effect size): " %5.3f `cohens_d'

Cohenovo d (effect size): 0.587

310 . display "Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt"
Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt

311 .
312 . * --- 5.2 ANOVA: Liší se platy mezi AI tiers? ---

```

```

313 . * Testuje rozdily mezi none, ai_integration, ai_focused
314 .
315 . display _n "--- 5.2 ANOVA: Plat podle AI tier ---"

```

--- 5.2 ANOVA: Plat podle AI tier ---

```

316 . oneway salary_mid ai_tier_num, tabulate bonferroni

```

ai_tier_num	Summary of Rocni plat - stredni hodnota (USD)		
	Mean	Std. Dev.	Freq.
ai_integr	140487.23	46483.006	1,946
applied_a	150497.53	48356.795	1,094
none	119532.29	39070.474	11,600
Total	124631.62	42133.036	14,640

Source	Analysis of Variance			F	Prob > F
	SS	df	MS		
Between groups	1.5228e+12	2	7.6140e+11	455.55	0.0000
Within groups	2.4464e+13	14637	1.6714e+09		
Total	2.5987e+13	14639	1.7752e+09		

Bartlett's test for equal variances: $\chi^2(2) = 185.3472$ Prob> $\chi^2 = 0.000$

Comparison of Rocni plat - stredni hodnota (USD) by ai_tier_num
(Bonferroni)

Row Mean- Col Mean	ai_integ	applied_
applied_	10010.3 0.000	
none	-20954.9 0.000	-30965.2 0.000

```

317 .
318 . * --- 5.3 Chi-square: Vzdelání a AI požadavky ---
319 . * Jsou AI pozice náročnější na vzdělání?
320 .
321 . display _n "--- 5.3 Chi-square: Vzdelani x AI tier ---"

```

--- 5.3 Chi-square: Vzdelani x AI tier ---

```

322 . tab edu_cat has_ai if edu_cat > 0, chi2 expected

```

Key
<i>frequency</i>
<i>expected frequency</i>

Pozadovane vzdelani (Binary)	has_ai		Total
	0	1	
Bachelor or Higher	8,876	1,895	10,771
	8,876.0	1,895.0	10,771.0
Total	8,876	1,895	10,771
	8,876.0	1,895.0	10,771.0

```

323 .
324 . * --- 5.4 Chi-square: Zkušenosti a AI požadavky ---
325 . display _n "--- 5.4 Chi-square: Zkusenosti x AI tier ---"

```

--- 5.4 Chi-square: Zkusenosti x AI tier ---

```

326 . tab exp_category has_ai, chi2 expected

```

Key
<i>frequency</i>
<i>expected frequency</i>

Kategorie seniority	has_ai		Total
	0	1	
Missing	2,169 2,086.2	422 504.8	2,591 2,591.0
Junior (0-2)	2,688 2,587.1	525 625.9	3,213 3,213.0
Mid (3-5)	6,650 6,768.4	1,756 1,637.6	8,406 8,406.0
Senior+ (6+)	2,864 2,929.3	774 708.7	3,638 3,638.0
Total	14,371 14,371.0	3,477 3,477.0	17,848 17,848.0

Pearson $\chi^2(3) = 55.1641$ Pr = 0.000

```

327 .
328 . * --- 5.5 Mann-Whitney U test (neparametrický) ---
329 . * Pro případ, že platy nemají normální rozdělení
330 .
331 . display _n "--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---"

```

--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---

```

332 . ranksum salary_mid, by(has_ai)

```

Two-sample Wilcoxon rank-sum (Mann-Whitney) test

has_ai	obs	rank sum	expected
0	11765	80679688	86125683
1	2875	26492432	21046438
combined	14640	1.072e+08	1.072e+08

unadjusted variance **4.127e+10**

adjustment for ties **-619148.58**

adjusted variance **4.127e+10**

```
Ho: salary~d(has_ai==0) = salary~d(has_ai==1)
      z = -26.808
      Prob > |z| = 0.0000
```

```
333 .
334 .
335 . * =====
336 . * 6. REGRESNÍ ANALÝZA – MODELY VEDOUCÍHO
337 . * =====
338 . * log(mzda) = skill cluster + AI tier + sektor + lokace + remote + typ + velikost
339 . *               + (vzdělání, zkušenosti, pozice)
340 . * Závorka = Model B (plný), bez závorky = Model A (base)
341 .
342 . display _n "=====
```

=====

```
343 . display "6. REGRESNI ANALYZA – MODELY VEDOUCIHO"
      6. REGRESNI ANALYZA – MODELY VEDOUCIHO
```

=====

```
344 . display "=====
```

=====

```
345 .
346 . * --- 6.0 Příprava: log transformace platu ---
347 . gen ln_salary = ln(salary_mid)
      (3,208 missing values generated)
```

```
348 . label variable ln_salary "Prirozeny logaritmus platu"
```

```
349 .
350 . summarize ln_salary, detail
```

Prirozeny logaritmus platu				
	Percentiles	Smallest		
1%	10.91509	9.903487		
5%	11.15625	10.27918		
10%	11.27649	10.30895	Obs	14,640
25%	11.46163	10.30895	Sum of Wgt.	14,640

50%	11.67746		Mean	11.68007
		Largest	Std. Dev.	.3253692
75%	11.90319	13.00583		
90%	12.10071	13.00583	Variance	.1058651
95%	12.20607	13.00583	Skewness	-.0110062
99%	12.40595	13.017	Kurtosis	3.390068

```
351 . display _n "Pozn: koeficient b v log modelu = (exp(b)-1)*100 % zmena platu"
```

Pozn: koeficient b v log modelu = (exp(b)-1)*100 % zmena platu

```
352 .
353 . * --- 6.1 Model A: Base model (bez individuálních charakteristik) ---
354 . * DV: ln(salary_mid)
355 . * IV: 24 skill clusterů + AI level + sektor NACE + region + remote + typ + velikost
356 .
357 . display _n "--- 6.1 Model A: Base model ---"
```

--- 6.1 Model A: Base model ---

```
358 . display "ln(plat) = skill clustery + AI level + sektor + region + remote + typ + velikost"
ln(plat) = skill clustery + AI level + sektor + region + remote + typ + velikost
```

```
359 .
360 . regress ln_salary ///
>   cluster_* ///
>   i.ai_level ///
>   i.sector_nace_num ///
>   i.region_num ///
>   is_remote ///
>   i.type_cat ///
>   i.size_cat ///
>   if ln_salary != ., vce(robust)
```

Linear regression	Number of obs	=	14,640
	F(44, 14595)	=	121.38
	Prob > F	=	0.0000
	R-squared	=	0.2469
	Root MSE	=	.28279

ln_salary	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
cluster_architecture__methods	.043874	.0069074	6.35	0.000	.0303346	.0574134
cluster_bi__analytics	-.0336277	.0050524	-6.66	0.000	-.0435311	-.0237243
cluster_backend_development	.0047283	.0057624	0.82	0.412	-.0065667	.0160233
cluster_certifications	.0058569	.0094537	0.62	0.536	-.0126736	.0243875
cluster_cloud_computing	.0552608	.0057825	9.56	0.000	.0439265	.0665952
cluster_data_engineering	.0220469	.005558	3.97	0.000	.0111524	.0329414
cluster_data_science__ml	.0104008	.0091764	1.13	0.257	-.0075862	.0283877
cluster_databases__storage	-.0373652	.0054673	-6.83	0.000	-.0480819	-.0266486
cluster_devops__containers	.0558644	.0055636	10.04	0.000	.0449591	.0667698
cluster_dynamic__web	.0136521	.0054104	2.52	0.012	.003047	.0242572
cluster_enterprise__managed	-.0208548	.0053904	-3.87	0.000	-.0314207	-.0102888
cluster_enterprise_platforms	-.0069907	.011099	-0.63	0.529	-.028746	.0147647
cluster_frontend_development	.0019515	.0055518	0.35	0.725	-.0089307	.0128336
cluster_generative_ai	.0408929	.0112935	3.62	0.000	.0187562	.0630296
cluster_mobile__desktop	-.0098489	.0068441	-1.44	0.150	-.0232643	.0035664
cluster_networking	.006598	.0055028	1.20	0.231	-.0041882	.0173841
cluster_os__embedded	-.0019257	.006348	-0.30	0.762	-.0143686	.0105173
cluster_scripting__shell	-.0108479	.0078752	-1.38	0.168	-.0262843	.0045884
cluster_security__identity	.0249826	.0053246	4.69	0.000	.0145458	.0354194
cluster_systems_programming	.0901649	.0069446	12.98	0.000	.0765525	.1037772
cluster_testing_qa__debugging	-.0177983	.0049824	-3.57	0.000	-.0275645	-.0080322
ai_level						
AI Integration	.0899703	.0084123	10.70	0.000	.0734812	.1064594
Applied/Core AI	.1276297	.0118683	10.75	0.000	.1043665	.150893
sector_nace_num						
J	.0517087	.0071508	7.23	0.000	.0376921	.0657252
K	.0201018	.0092324	2.18	0.029	.002005	.0381985
M	.0549893	.0130778	4.20	0.000	.0293551	.0806234
Other	-.0413269	.008934	-4.63	0.000	-.0588387	-.023815
Q	.0102256	.0130206	0.79	0.432	-.0152964	.0357475
Unknown	.0223882	.010034	2.23	0.026	.0027204	.0420561
region_num						
Northeast	.1075579	.0087335	12.32	0.000	.0904392	.1246766
South	.0666334	.0072231	9.22	0.000	.0524751	.0807917
Unknown	.1939568	.0100235	19.35	0.000	.1743096	.2136041
West	.2406722	.0077666	30.99	0.000	.2254487	.2558956
is_remote	.0672041	.0070535	9.53	0.000	.0533783	.0810299
type_cat						

Private/Subsidiary	.032117	.0112727	2.85	0.004	.010021	.054213
Public	.0433775	.0116631	3.72	0.000	.0205163	.0662387
Nonprofit/Gov/Edu	-.039119	.0159748	-2.45	0.014	-.0704317	-.0078064
size_cat						
1-50	-.0047936	.0094284	-0.51	0.611	-.0232745	.0136873
51-200	.006199	.0104961	0.59	0.555	-.0143747	.0267727
201-500	.0046206	.0111613	0.41	0.679	-.017257	.0264982
501-1000	.0391504	.0125224	3.13	0.002	.0146049	.0636959
1001-5000	.0641218	.0104837	6.12	0.000	.0435724	.0846712
5001-10000	.090369	.0131435	6.88	0.000	.0646062	.1161319
10000+	.1207694	.0101091	11.95	0.000	.1009543	.1405844
_cons	11.33169	.0154997	731.09	0.000	11.30131	11.36207

```

361 .
362 . estimates store model_a

363 . display _n "Model A: R2 = " e(r2) " , Adj R2 = " e(r2_a) " , N = " e(N)

```

Model A: R2 = .24688373, Adj R2 = .24461329, N = 14640

```

364 .
365 . * --- 6.1b VIF diagnostika (kolinearita) ---
366 . display _n "--- 6.1b VIF diagnostika Model A ---"

```

--- 6.1b VIF diagnostika Model A ---

```

367 . quietly regress ln_salary ///
> cluster_* ///
> i.ai_level ///
> i.sector_nace_num ///
> i.region_num ///
> is_remote ///
> i.type_cat ///
> i.size_cat ///
> if ln_salary != .

```

368 . vif

Variable	VIF	1/VIF
cluster_ar~s	1.17	0.853898
cluster_bi~s	1.08	0.927379
cluster_ba~t	1.44	0.696803
cluster_ce~s	1.09	0.920250
cluster_cl~g	1.47	0.679847
cluster_da~g	1.23	0.811149
cluster_da~l	1.53	0.651484
cluster_da~e	1.35	0.739426
cluster_de~s	1.42	0.704143
cluster_dy~b	1.30	0.768172
cluster_en~d	1.26	0.792424
cluster_en~s	1.03	0.970279
cluster_fr~t	1.35	0.741251
cluster_ge~i	1.28	0.780858
cluster_mo~p	1.14	0.874584
cluster_ne~g	1.13	0.884530
cluster_os~d	1.38	0.726264
cluster_sc~l	1.31	0.761808
cluster_se~y	1.22	0.822830
cluster_sy~g	1.22	0.818938
cluster_te~g	1.11	0.904606
ai_level		
1	1.31	0.762462
2	1.70	0.587763
sector_nac~m		
2	2.45	0.407443
3	1.55	0.643940
4	1.29	0.775334
5	1.90	0.524993
6	1.33	0.751854
7	2.91	0.343261
region_num		
2	1.84	0.543625
3	2.52	0.397183
4	2.62	0.381998
5	2.43	0.411986
is_remote	1.63	0.612834
type_cat		
1	5.09	0.196532
2	4.82	0.207481
4	2.12	0.472729
size_cat		
1	2.07	0.482771

2	2.06	0.486466
3	1.78	0.561146
4	1.60	0.626809
5	2.20	0.455061
6	1.67	0.597703
7	3.14	0.318308
Mean VIF	1.79	

```

369 .
370 . * --- 6.2 Model B: Plný model (+vzdělání, zkušenosti, pozice) ---
371 . display _n "--- 6.2 Model B: Plný model ---"

--- 6.2 Model B: Plný model ---

372 . display "ln(plat) = Model A + vzdelani + zkusenosti + pozice (job family)"
    ln(plat) = Model A + vzdelani + zkusenosti + pozice (job family)

373 .
374 . regress ln_salary ///
    > cluster_* ///
    > i.ai_level ///
    > i.sector_nace_num ///
    > i.region_num ///
    > is_remote ///
    > i.type_cat ///
    > i.size_cat ///
    > i.edu_cat ///
    > i.exp_category ///
    > i.job_family_num ///
    > if ln_salary != ., vce(robust)

```

Linear regression	Number of obs	=	14,640
	F(54, 14585)	=	195.85
	Prob > F	=	0.0000
	R-squared	=	0.3762
	Root MSE	=	.25745

ln_salary	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
cluster_architecture__methods	.0182767	.0063822	2.86	0.004	.0057667	.0307866
cluster_bi__analytics	-.0245583	.0046442	-5.29	0.000	-.0336616	-.015455
cluster_backend_development	-.0086889	.0053437	-1.63	0.104	-.0191633	.0017855
cluster_certifications	-.0200132	.0086134	-2.32	0.020	-.0368965	-.0031299
cluster_cloud_computing	.0258901	.0053521	4.84	0.000	.0153993	.0363809
cluster_data_engineering	.0153715	.0051162	3.00	0.003	.0053431	.0254
cluster_data_science_ml	.0166477	.0085328	1.95	0.051	-.0000777	.0333731
cluster_databases__storage	-.0359948	.0050603	-7.11	0.000	-.0459138	-.0260759
cluster_devops__containers	.025118	.0051685	4.86	0.000	.014987	.035249
cluster_dynamic__web	.0267062	.0049901	5.35	0.000	.016925	.0364873
cluster_enterprise__managed	-.018417	.0050056	-3.68	0.000	-.0282286	-.0086054
cluster_enterprise_platforms	-.0116828	.0097758	-1.20	0.232	-.0308446	.007479
cluster_frontend_development	.000914	.0051075	0.18	0.858	-.0090973	.0109253
cluster_generative_ai	.026495	.0103596	2.56	0.011	.006189	.0468011
cluster_mobile__desktop	-.0067986	.0062776	-1.08	0.279	-.0191036	.0055063
cluster_networking	.0101547	.0050489	2.01	0.044	.0002582	.0200512
cluster_os__embedded	.0032115	.0058095	0.55	0.580	-.0081758	.0145988
cluster_scripting__shell	-.0144299	.0073313	-1.97	0.049	-.0288002	-.0000595
cluster_security__identity	.0228614	.0049397	4.63	0.000	.013179	.0325438
cluster_systems_programming	.0735347	.0065292	11.26	0.000	.0607366	.0863328
cluster_testing_qa__debugging	-.0177245	.004559	-3.89	0.000	-.0266608	-.0087882
ai_level						
AI Integration	.0772525	.007672	10.07	0.000	.0622145	.0922905
Applied/Core AI	.0987976	.0111026	8.90	0.000	.0770351	.12056
sector_nace_num						
J	.0442937	.0064729	6.84	0.000	.031606	.0569813
K	.011515	.008302	1.39	0.165	-.0047579	.0277879
M	.0622706	.0120566	5.16	0.000	.0386382	.0859029
Other	-.0440756	.0082282	-5.36	0.000	-.0602038	-.0279474
Q	.0006889	.0112156	0.06	0.951	-.0212951	.0226729
Unknown	.0162056	.0092557	1.75	0.080	-.0019367	.0343479
region_num						
Northeast	.1020685	.0078468	13.01	0.000	.0866878	.1174492
South	.0555782	.0064622	8.60	0.000	.0429114	.0682449
Unknown	.1675965	.0091078	18.40	0.000	.149744	.185449
West	.225656	.0069962	32.25	0.000	.2119426	.2393694
is_remote	.0353068	.0064487	5.48	0.000	.0226666	.047947
type_cat						

Private/Subsidiary	.0389782	.0102758	3.79	0.000	.0188364	.05912
Public	.0480561	.0106697	4.50	0.000	.0271421	.06897
Nonprofit/Gov/Edu	-.0075057	.0150237	-0.50	0.617	-.036954	.0219426
size_cat						
1-50	-.0116666	.0086453	-1.35	0.177	-.0286125	.0052794
51-200	-.0015768	.0096148	-0.16	0.870	-.020423	.0172694
201-500	-.0071054	.0101763	-0.70	0.485	-.0270522	.0128413
501-1000	.0211831	.0111515	1.90	0.058	-.0006752	.0430415
1001-5000	.0426998	.0096786	4.41	0.000	.0237285	.0616712
5001-10000	.0752297	.0121559	6.19	0.000	.0514025	.0990569
10000+	.1122612	.0093831	11.96	0.000	.0938691	.1306532
edu_cat						
Bachelor or Higher	-.0181339	.0047531	-3.82	0.000	-.0274506	-.0088172
exp_category						
Junior (0-2)	-.1776444	.008284	-21.44	0.000	-.1938821	-.1614066
Mid (3-5)	-.0006716	.0073433	-0.09	0.927	-.0150654	.0137223
Senior+ (6+)	.1206672	.0082501	14.63	0.000	.104496	.1368383
job_family_num						
DevOps & Cloud	.0694456	.0127847	5.43	0.000	.0443859	.0945053
Management	.0922822	.0136606	6.76	0.000	.0655057	.1190586
Other	-.0445414	.0110483	-4.03	0.000	-.0661975	-.0228854
Software Developer	-.058153	.0115376	-5.04	0.000	-.0807681	-.0355378
Software Engineer	.0699598	.0122118	5.73	0.000	.0460231	.0938965
Sr+ Software Engineer	.1712239	.0123	13.92	0.000	.1471144	.1953333
_cons	11.43486	.0185508	616.41	0.000	11.3985	11.47122

375 .

376 . estimates store model_b

```
377 . display _n "Model B: R2 = " e(r2) ", Adj R2 = " e(r2_a) ", N = " e(N)
```

Model B: R2 = .37623628, Adj R2 = .37392684, N = 14640

```
378 .
379 . * --- 6.2b VIF diagnostika Model B ---
380 . display _n "--- 6.2b VIF diagnostika Model B ---"
```

--- 6.2b VIF diagnostika Model B ---

```
381 . quietly regress ln_salary ///
>   cluster_* ///
>   i.ai_level ///
>   i.sector_nace_num ///
>   i.region_num ///
>   is_remote ///
>   i.type_cat ///
>   i.size_cat ///
>   i.edu_cat ///
>   i.exp_category ///
>   i.job_family_num ///
>   if ln_salary != .
```

```
382 . vif
```

Variable	VIF	1/VIF
cluster_ar~s	1.19	0.837683
cluster_bi~s	1.10	0.908104
cluster_ba~t	1.49	0.671303
cluster_ce~s	1.10	0.906538
cluster_cl~g	1.51	0.662101
cluster_da~g	1.27	0.788284
cluster_da~l	1.55	0.645821
cluster_da~e	1.40	0.712630
cluster_de~s	1.49	0.672416
cluster_dy~b	1.33	0.754659
cluster_en~d	1.29	0.775647
cluster_en~s	1.03	0.966434
cluster_fr~t	1.38	0.725688
cluster_ge~i	1.29	0.778166
cluster_mo~p	1.15	0.869984
cluster_ne~g	1.14	0.876750
cluster_os~d	1.39	0.720293
cluster_sc~l	1.37	0.727499
cluster_se~y	1.25	0.802326
cluster_sy~g	1.26	0.793045

cluster_te~g	1.12	0.895962
ai_level		
1	1.32	0.758012
2	1.74	0.573427
sector_nac~m		
2	2.49	0.401713
3	1.56	0.641016
4	1.29	0.773804
5	1.91	0.522423
6	1.33	0.750453
7	2.95	0.339384
region_num		
2	1.84	0.542905
3	2.53	0.396000
4	2.63	0.380250
5	2.44	0.410306
is_remote	1.66	0.601807
type_cat		
1	5.09	0.196372
2	4.82	0.207323
4	2.12	0.470724
size_cat		
1	2.07	0.482138
2	2.06	0.485492
3	1.79	0.559133
4	1.60	0.623473
5	2.22	0.449987
6	1.69	0.592621
7	3.18	0.314423
1.edu_cat	1.14	0.877175
exp_category		
2	2.01	0.497720
3	2.49	0.402243
4	2.22	0.451402
job_family~m		
2	2.92	0.342883
3	1.86	0.538105
4	6.52	0.153292
5	4.39	0.227575
6	3.36	0.297352
7	2.47	0.405059
Mean VIF	2.02	

```
383 .
384 . * --- 6.3 Srovnání modelů A vs B (F-test nested models) ---
385 . display _n "--- 6.3 Srovnání modelu A vs B ---"

    --- 6.3 Srovnání modelu A vs B ---

386 . display "F-test: testuje zda edu + exp + job_family významně zlepšuje model"
    F-test: testuje zda edu + exp + job_family významně zlepšuje model

387 .
388 . quietly regress ln_salary ///
    > cluster_* ///
    > i.ai_level ///
    > i.sector_nace_num ///
    > i.region_num ///
    > is_remote ///
    > i.type_cat ///
    > i.size_cat ///
    > if ln_salary != .

389 . estimates store model_a_lr

390 .
391 . quietly regress ln_salary ///
    > cluster_* ///
    > i.ai_level ///
    > i.sector_nace_num ///
    > i.region_num ///
    > is_remote ///
    > i.type_cat ///
    > i.size_cat ///
    > i.edu_cat ///
    > i.exp_category ///
    > i.job_family_num ///
    > if ln_salary != .
```

```
392 . estimates store model_b_lr
```

```
393 .
```

```
394 . lrtest model_a_lr model_b_lr
```

```

Likelihood-ratio test                                LR chi2(10) =    2758.88
(Assumption: model_a_lr nested in model_b_lr)      Prob > chi2 =    0.0000

```

```
395 . estimates drop model_a_lr model_b_lr
```

```
396 .
```

```
397 . * --- 6.4 Porovnávací tabulka modelů ---
```

```
398 . display _n "---- 6.4 Porovnani koeficientu Model A vs B ----"
```

--- 6.4 Porovnani koeficientu Model A vs B ---

```
399 . estimates table model_a model_b, star stats(N r2 r2_a)
```

Variable	model_a	model_b
cluster_ar~s	.04387401***	.01827666**
cluster_bi~s	-.03362769***	-.02455831***
cluster_ba~t	.00472829	-.00868892
cluster_ce~s	.00585694	-.02001323*
cluster_cl~g	.05526085***	.02589009***
cluster_da~g	.02204689***	.01537153**
cluster_da~l	.01040078	.0166477
cluster_da~e	-.03736524***	-.03599484***
cluster_de~s	.05586444***	.02511798***
cluster_dy~b	.01365212*	.02670616***
cluster_en~d	-.02085478***	-.01841702***
cluster_en~s	-.00699066	-.01168283
cluster_fr~t	.00195148	.00091402
cluster_ge~i	.04089292***	.02649504*
cluster_mo~p	-.00984895	-.00679864
cluster_ne~g	.00659796	.0101547*
cluster_os~d	-.00192567	.00321151
cluster_sc~l	-.01084794	-.01442986*
cluster_se~y	.02498256***	.02286135***
cluster_sy~g	.09016487***	.0735347***
cluster_te~g	-.01779832***	-.01772447***
ai_level		
AI Integr..	.0899703***	.07725247***
Applied/C..	.12762974***	.09879757***

sector_nac~m		
J	.05170866***	.04429368***
K	.02010175*	.01151497
M	.05498929***	.06227056***
Other	-.04132689***	-.04407559***
Q	.01022558	.00068889
Unknown	.02238824*	.0162056
region_num		
Northeast	.10755788***	.10206851***
South	.06663341***	.05557817***
Unknown	.19395684***	.16759651***
West	.24067218***	.22565597***
is_remote	.06720409***	.0353068***
type_cat		
Private/S~y	.03211699**	.03897818***
Public	.04337748***	.04805606***
Nonprofit~u	-.03911905*	-.00750567
size_cat		
1-50	-.00479358	-.01166655
51-200	.00619899	-.0015768
201-500	.0046206	-.00710542
501-1000	.0391504**	.02118313
1001-5000	.06412177***	.04269982***
5001-10000	.09036903***	.07522974***
10000+	.12076939***	.11226116***
edu_cat		
Bachelor ..		-.01813394***
exp_category		
Junior (..)		-.17764436***
Mid (3-5)		-.00067156
Senior+ ..)		.12066715***
job_family~m		
DevOps & ..		.06944558***
Management		.09228219***
Other		-.04454144***
Software ..		-.05815298***
Software ..		.06995981***
Sr+ Softw..		.17122386***
_cons	11.331689***	11.434862***

salary_mid	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
has_ai	21473.41	907.5946	23.66	0.000	19694.41	23252.41
edu_cat No Degree / Missing	3116.547	686.1418	4.54	0.000	1771.622	4461.471
exp_category Missing	-2876.471	1121.908	-2.56	0.010	-5075.553	-677.3892
Junior (0-2)	-20235.3	778.9522	-25.98	0.000	-21762.14	-18708.45
Senior+ (6+)	19219.61	863.5728	22.26	0.000	17526.9	20912.32
is_remote	11011.55	835.1548	13.19	0.000	9374.54	12648.56
_cons	116988.9	549.5954	212.86	0.000	115911.6	118066.2

```

413 .
414 . * --- 6B.2 Logistická regrese: Prediktory AI požadavků ---
415 . display _n "--- 6B.2 Logisticka regrese: Prediktory AI pozadavku ---"

```

```

--- 6B.2 Logisticka regrese: Prediktory AI pozadavku ---

```

```

416 . logit has_ai ib1.edu_cat ib3.exp_category is_remote, or

```

```

Iteration 0: log likelihood = -8801.3004
Iteration 1: log likelihood = -8677.4818
Iteration 2: log likelihood = -8676.092
Iteration 3: log likelihood = -8676.0919

```

```

Logistic regression      Number of obs      =      17,848
                        LR chi2(5)          =      250.42
                        Prob > chi2         =      0.0000
Log likelihood = -8676.0919  Pseudo R2          =      0.0142

```

has_ai	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
edu_cat No Degree / Missing	1.287564	.0506896	6.42	0.000	1.19195	1.390847
exp_category Missing	.7235071	.043823	-5.34	0.000	.642518	.8147049
Junior (0-2)	.806529	.0446531	-3.88	0.000	.7235918	.8989724
Senior+ (6+)	1.012611	.0495351	0.26	0.798	.9200329	1.114505
is_remote	1.586294	.064628	11.33	0.000	1.46455	1.718157
_cons	.2032415	.0069457	-46.62	0.000	.1900741	.2173212

Note: **_cons** estimates baseline odds.

417 . display _n "--- 6B.2b Marginalni efekty ---"

--- 6B.2b Marginalni efekty ---

418 . margins, dydx(*) atmeans

Conditional marginal effects Number of obs = **17,848**
Model VCE : **OIM**

Expression : **Pr(has_ai), predict()**
dy/dx w.r.t. : **0.edu_cat 0.exp_category 2.exp_category 4.exp_category is_remote**
at : 0.edu_cat = **.396515** (mean)
 1.edu_cat = **.603485** (mean)
 0.exp_cate~y = **.1451703** (mean)
 2.exp_cate~y = **.1800202** (mean)
 3.exp_cate~y = **.4709771** (mean)
 4.exp_cate~y = **.2038324** (mean)
 is_remote = **.2838974** (mean)

	Delta-method					
	dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
edu_cat No Degree / Missing	.0396507	.0062604	6.33	0.000	.0273806	.0519208
exp_category Missing	-.0475614	.00839	-5.67	0.000	-.0640055	-.0311174
Junior (0-2)	-.0326938	.0081476	-4.01	0.000	-.0486627	-.0167249
Senior+ (6+)	.0020417	.0079815	0.26	0.798	-.0136017	.0176852

is_remote	.0712175	.0062486	11.40	0.000	.0589704	.0834646
-----------	----------	----------	-------	-------	----------	----------

Note: dy/dx for factor levels is the discrete change from the base level.

```

419 .
420 . * --- 6B.3 Multinomialni logit ---
421 . display _n "--- 6B.3 Multinomialni logit: Uroven AI pozadavku ---"

```

--- 6B.3 Multinomialni logit: Uroven AI pozadavku ---

```

422 . mlogit ai_level ib1.edu_cat ib3.exp_category is_remote, baseoutcome(0) rrr

```

```

Iteration 0: log likelihood = -11456.516
Iteration 1: log likelihood = -11316.046
Iteration 2: log likelihood = -11314.341
Iteration 3: log likelihood = -11314.341

```

Multinomial logistic regression	Number of obs	=	17,848
	LR chi2(10)	=	284.35
	Prob > chi2	=	0.0000
Log likelihood = -11314.341	Pseudo R2	=	0.0124

ai_level	RRR	Std. Err.	z	P> z	[95% Conf. Interval]	
None	(base outcome)					
AI_Integration						
edu_cat						
No Degree / Missing	1.406013	.0648235	7.39	0.000	1.284532	1.538982
exp_category						
Missing	.6266465	.0458402	-6.39	0.000	.542945	.7232516
Junior (0-2)	.7632281	.0498374	-4.14	0.000	.6715409	.8674336
Senior+ (6+)	.9275572	.0537132	-1.30	0.194	.8280358	1.03904
is_remote	1.515232	.0726074	8.67	0.000	1.379402	1.664437
_cons	.1424789	.005707	-48.65	0.000	.1317211	.1541153
Applied_Core_AI						
edu_cat						
No Degree / Missing	1.116677	.0672701	1.83	0.067	.9923164	1.256624
exp_category						
Missing	.8828497	.0794959	-1.38	0.166	.7400151	1.053254
Junior (0-2)	.8621776	.0734988	-1.74	0.082	.7295139	1.018967
Senior+ (6+)	1.164495	.0849232	2.09	0.037	1.009397	1.343424

is_remote	1.679625	.1029225	8.46	0.000	1.489544	1.893963
_cons	.0757693	.003996	-48.92	0.000	.0683284	.0840205

Note: **_cons** estimates baseline relative risk for each outcome.

423 . display _n "---- 6B.3b Marginalni efekty: P(AI Integration) ----"

--- 6B.3b Marginalni efekty: P(AI Integration) ---

424 . margins, dydx(*) predict(outcome(1)) atmeans

Conditional marginal effects Number of obs = 17,848
Model VCE : OIM

Expression : Pr(ai_level==AI_Integration), predict(outcome(1))
dy/dx w.r.t. : 0.edu_cat 0.exp_category 2.exp_category 4.exp_category is_remote
at : 0.edu_cat = .396515 (mean)
 1.edu_cat = .603485 (mean)
 0.exp_cate~y = .1451703 (mean)
 2.exp_cate~y = .1800202 (mean)
 3.exp_cate~y = .4709771 (mean)
 4.exp_cate~y = .2038324 (mean)
 is_remote = .2838974 (mean)

	Delta-method				[95% Conf. Interval]	
	dy/dx	Std. Err.	z	P> z		
edu_cat						
No Degree / Missing	.0381397	.0053799	7.09	0.000	.0275954	.048684
exp_category						
Missing	-.0477517	.0068776	-6.94	0.000	-.0612315	-.0342718
Junior (0-2)	-.0290197	.0069593	-4.17	0.000	-.0426597	-.0153797
Senior+ (6+)	-.0106123	.0067663	-1.57	0.117	-.0238741	.0026494
is_remote	.0417334	.0052911	7.89	0.000	.0313629	.0521038

Note: dy/dx for factor levels is the discrete change from the base level.

425 . display _n "--- 6B.3c Marginalni efekty: P(Applied/Core AI) ---"

--- 6B.3c Marginalni efekty: P(Applied/Core AI) ---

426 . margins, dydx(*) predict(outcome(2)) atmeans

Conditional marginal effects Number of obs = 17,848
Model VCE : OIM

Expression : Pr(ai_level==Applied_Core_AI), predict(outcome(2))
dy/dx w.r.t. : 0.edu_cat 0.exp_category 2.exp_category 4.exp_category is_remote
at
 : 0.edu_cat = .396515 (mean)
 : 1.edu_cat = .603485 (mean)
 : 0.exp_cate~y = .1451703 (mean)
 : 2.exp_cate~y = .1800202 (mean)
 : 3.exp_cate~y = .4709771 (mean)
 : 4.exp_cate~y = .2038324 (mean)
 : is_remote = .2838974 (mean)

	Delta-method		z	P> z	[95% Conf. Interval]	
	dy/dx	Std. Err.				
edu_cat No Degree / Missing	.0041446	.004049	1.02	0.306	-.0037913	.0120805
exp_category Missing	-.0042175	.0057276	-0.74	0.462	-.0154434	.0070084
Junior (0-2)	-.0070592	.005279	-1.34	0.181	-.0174058	.0032874
Senior+ (6+)	.0117128	.0053184	2.20	0.028	.001289	.0221367
is_remote	.0309173	.0040216	7.69	0.000	.0230352	.0387994

Note: dy/dx for factor levels is the discrete change from the base level.

```
427 .
428 . * Hausman test IIA
429 . display _n "---- 6B.3d Hausman test IIA ----"

    --- 6B.3d Hausman test IIA ---

430 . quietly mlogit ai_level ib1.edu_cat ib3.exp_category is_remote, baseoutcome(0)
431 . estimates store full_model
432 . quietly mlogit ai_level ib1.edu_cat ib3.exp_category is_remote if ai_level != 1, baseoutcome(0)
433 . estimates store reduced_model
434 . capture hausman reduced_model full_model, alleqs constant
435 . if _rc {
436 .     display "Hausman test nelze provest (typicke pro male vzorky v nekterych kategoriich)"
437 . }

438 . estimates drop full_model reduced_model

439 .
440 .
441 . * =====
442 . * 7. ANALÝZA HARD SKILLS
443 . * =====
444 . * Rozbor nejčastějších technických dovedností
445 .
446 . display _n "=====
=====

447 . display "7. ANALYZA HARD SKILLS"
    7. ANALYZA HARD SKILLS
```

```

448 . display "=====
=====

449 .
450 . * Poznámka: hardskills je textový sloupec s čárkami oddělenými skills
451 . * Pro detailní analýzu je lepší použít Python k vytvoření dummy proměnných
452 . * Zde ukážeme základní exploraci
453 .
454 . * Počet unique skills na pozici (aproximace)
455 . gen skill_count = 1 + length(hardskills) - length(subinstr(hardskills, ",", "", .))

456 . replace skill_count = 0 if hardskills == ""
    (110 real changes made)

457 . label variable skill_count "Pocet pozadovanych hard skills"

458 .
459 . display _n "--- 7.1 Pocet skills na pozici ---"

    --- 7.1 Pocet skills na pozici ---

460 . summarize skill_count, detail

```

Pocet pozadovanych hard skills

	Percentiles	Smallest		
1%	1	0		
5%	3	0		
10%	5	0	Obs	17,848
25%	9	0	Sum of Wgt.	17,848
50%	15		Mean	16.78737
		Largest	Std. Dev.	10.30623
75%	23	72		
90%	31	72	Variance	106.2184
95%	36	80	Skewness	.9651727
99%	47	80	Kurtosis	4.326039

```
461 . tabstat skill_count, by(desc_tier_llm) statistics(count mean sd min max)
```

Summary for variables: skill_count
by categories of: desc_tier_llm

desc_tier_llm	N	mean	sd	min	max
ai_integration	2354	19.46432	10.3817	0	68
applied_ai	1313	20.5834	10.80076	1	80
none	14181	15.99154	10.09214	0	80
Total	17848	16.78737	10.30623	0	80

```
462 .
463 . * T-test: Vyžadují AI pozice více skills?
464 . display _n "--- 7.2 T-test: Pocet skills AI vs non-AI ---"
```

--- 7.2 T-test: Pocet skills AI vs non-AI ---

```
465 . ttest skill_count, by(has_ai)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
0	14,371	16.04704	.0843192	10.10811	15.88176	16.21232
1	3,477	19.84728	.1789442	10.55164	19.49644	20.19813
combined	17,848	16.78737	.0771445	10.30623	16.63616	16.93858
diff		-3.800243	.192699		-4.177952	-3.422534

diff = mean(0) - mean(1) t = **-19.7211**
Ho: diff = 0 degrees of freedom = **17846**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = **0.0000** Pr(|T| > |t|) = **0.0000** Pr(T > t) = **1.0000**

```

466 .
467 .
468 . * =====
469 . * 8. EXPORTY PRO TABULKY A GRAFY
470 . * =====
471 . * Příprava dat pro publikovatelné tabulky
472 .
473 . display _n "=====
=====

474 . display "8. EXPORTY"
      8. EXPORTY

475 . display "=====
=====

476 .
477 . * --- 8.1 Summary statistics ---
478 . display _n "--- 8.1 Summary statistics pro tabulky ---"

      --- 8.1 Summary statistics pro tabulky ---

479 . summarize salary_mid experience_min_llm edu_cat exp_category skill_count has_ai is_remote

```

Variable	Obs	Mean	Std. Dev.	Min	Max
salary_mid	14,640	124631.6	42133.04	20000	450000
experience~m	15,257	4.520433	2.625669	0	35
edu_cat	17,848	.603485	.4891874	0	1
exp_category	17,848	2.588301	1.233278	0	4
skill_count	17,848	16.78737	10.30623	0	80
has_ai	17,848	.1948117	.3960668	0	1
is_remote	17,848	.2838974	.4509003	0	1

```

480 .
481 . * Export do CSV pro další zpracování (grafy v Excelu/R/Python)
482 . export delimited desc_tier_llm salary_mid experience_min_llm edulevel_llm state sector using "$outdir/summary_for_charts.csv" if salary_mi
(note: file ./output/run__5_Mar_2026_13-50-35/summary_for_charts.csv not found)
file ./output/run__5_Mar_2026_13-50-35/summary_for_charts.csv saved

483 .
484 .
485 . * =====
486 . * 9. VIZUALIZACE
487 . * =====
488 . * Základní grafy pro diplomovou práci
489 .
490 . display _n "=====
=====

491 . display "9. VIZUALIZACE"
9. VIZUALIZACE

492 . display "=====
=====

493 .
494 . * --- 9.1 Sloupcový graf: AI tier distribuce ---
495 . graph bar (count), over(desc_tier_llm) ///
>     title("Distribuce AI požadavku v IT pozicích") ///
>     ytitle("Pocet pozic") ///
>     bar(1, color(navy))

496 . graph export "$outdir/ai_tier_distribution.png", replace width(1200)
(note: file ./output/run__5_Mar_2026_13-50-35/ai_tier_distribution.png not found)
(file ./output/run__5_Mar_2026_13-50-35/ai_tier_distribution.png written in PNG format)

497 .

```

```
498 . * --- 9.2 Box plot: Platy podle AI tier ---
499 . graph box salary_mid, over(desc_tier_llm) ///
>     title("Rozlozeni platu podle AI pozadavku") ///
>     ytitle("Rocni plat (USD)")

500 . graph export "$outdir/salary_by_ai_tier.png", replace width(1200)
(note: file ./output/run__5_Mar_2026_13-50-35/salary_by_ai_tier.png not found)
(file ./output/run__5_Mar_2026_13-50-35/salary_by_ai_tier.png written in PNG format)

501 .
502 . * --- 9.3 Histogram: Požadované zkušenosti ---
503 . histogram experience_min_llm if experience_min_llm < 15, ///
>     by(has_ai, title("Požadovane zkusenosti") note("")) ///
>     xtitle("Roky zkusenosti") percent bin(15)

504 . graph export "$outdir/experience_histogram.png", replace width(1200)
(note: file ./output/run__5_Mar_2026_13-50-35/experience_histogram.png not found)
(file ./output/run__5_Mar_2026_13-50-35/experience_histogram.png written in PNG format)

505 .
506 . * --- 9.4 Vzdělání podle AI tier ---
507 . graph bar (count), over(edu_cat) over(has_ai) ///
>     title("Vzdelavaci pozadavky: AI vs non-AI pozice") ///
>     ytitle("Pocet pozic") ///
>     legend(label(1 "non-AI") label(2 "AI pozice"))

508 . graph export "$outdir/education_by_ai.png", replace width(1200)
(note: file ./output/run__5_Mar_2026_13-50-35/education_by_ai.png not found)
(file ./output/run__5_Mar_2026_13-50-35/education_by_ai.png written in PNG format)

509 .
510 .
511 . * =====
512 . * 10. ZÁVĚR A ULOŽENÍ
513 . * =====
```

```
514 .
515 . display _n "=====
=====

516 . display "ANALYZA DOKONCENA"
ANALYZA DOKONCENA

517 . display "=====
=====

518 . display "Vystupy ulozeny do: $outdir"
Vystupy ulozeny do: ./output/run__5_Mar_2026_13-50-35

519 . display "Log soubor: $outdir/ai_skills_analysis_`time_string'.log"
Log soubor: ./output/run__5_Mar_2026_13-50-35/ai_skills_analysis__5_Mar_2026_13-50-35.log

520 .
521 . * Ulož zpracovaný dataset pro další práci
522 . save "$outdir/ai_skills_processed.dta", replace
    (note: file ./output/run__5_Mar_2026_13-50-35/ai_skills_processed.dta not found)
    file ./output/run__5_Mar_2026_13-50-35/ai_skills_processed.dta saved

523 .
524 . * Zavři log
525 . log close
      name: <unnamed>
      log: C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output/run__5_Mar_2026_13-50-35/ai_skills_analysis.log
      log type: text
      closed on: 5 Mar 2026, 14:53:18
```

```
526 .
527 . * =====
528 . * KONEC DO-FILU
529 . * =====
530 .
    end of do-file

531 .
```